Exercise for Sect. 2.9

Exercise **2.9.1**

Let $(X, \mathcal{B}, \mu, T)$ be a measure-preserving system, and ket $r : X \to \mathbb{N}_0$ be a map in L^1_μ. The suspension defined by r is the system $\left(X^{(r)}, \mathcal{B}^{(r)}, \mu^{(r)}, T^{(r)}\right)$, where

- $X^{(r)} = \{(x, n) : 0 \leq n < r(x)\}$;
- $\mathcal{B}^{(r)}$ is the product σ-algebra of $\mathcal{B}$ and the Borel σ-algebra on $\mathbb{N}$ (which comprise all subsets);
- $\mu^{(r)}$ is defined by $\mu^{(r)}(X \times N) = \frac{1}{\int r \, \mathrm{d}\mu} \mu(A) \times |N|$ for $A \in \mathcal{B}$ and $N \subseteq \mathbb{N}$;

and

- $$T^{(r)}(x, n) = \begin{cases} (x, n+1) & \text{if } n + 1 < r(x) \\ (T(x), 0) & \text{if } n + 1 = r(x). \end{cases}$$

1. Verify that this defines a finite measure-preserving system.
2. Show that the induced map on the set $A = \{(x, 0) : x \in X\}$ is isomorphic to the original system $(X, \mathcal{B}, \mu, T)$.

Proof: □

Exercise 2.9.2

An invertible measure-preserving system $(X, \mathcal{B}, \mu, T)$ is call aperidic if $\mu\big(\{x \in X : T^k(x) = x\}\big) = 0$ for all $k \in \mathbb{Z} \setminus \{0\}$.

1. Show that an ergodic transformation on a non-atomic space is aperiodic.
2. Find an example of an aperiodic transformation on a non-atomic space that is not ergodic.
3. Prove Kautani-Rokhlin lemma for an invertible aperiodic transformation on a non-atomic space.

Proof: $\square$

Exercise 2.9.3

Show that for an ergodic measure-preserving sustem $(X, \mathcal{B}, \mu, T)$, sequence $a_1, \ldots, a_n$ of distinct integers, and $\varepsilon > 0$ it is not always possible to find a measurable set A with the properties that $T^{a_1}(A), \ldots, T^{a_n}(A)$ are disjoint and $\mu\left(\bigcup_{i=1}^{n} T^{a_i}(A)\right) > \varepsilon$.

Proof: □

Exercise **2.9.4**

Let $(X, \mathcal{B}, \mu, T)$ be an invertible aperiodic measure-preserving system on a non-atomic space. Then, for any $\varepsilon > 0$, there is a set $A \in \mathcal{B}$ with $\mu(A) < \varepsilon$ with the property that for any finite set $F \subseteq X$, there is some $j = j(F)$ with $F \subseteq T^{-j}(A)$.

Proof: $\square$